

[University name]

**Air Pollution and Respiratory Morbidity in Kampala,
Uganda:
An Ecological Time-Series and Spatial Analysis of
Monthly Routine Surveillance Data (2020–2024)**

A thesis chapter submitted in partial fulfilment of the requirements for the
degree of *[PhD degree]*

[Author name]

Supervisors: *[Supervisor name(s)]*

[Department and university]

[September 2026]

Abstract

Background. Fine particulate air pollution (PM_{2.5}) presents an escalating public health crisis in rapidly urbanizing African metropolises. While daily and weekly surveillance capture acute exposure fluctuations, monthly aggregated surveillance data provide critical insight into sub-chronic disease dynamics, seasonal disease burden, and health-system capacity in resource-constrained settings. This chapter evaluates monthly associations between ambient PM_{2.5} and non-fatal respiratory morbidity (tuberculosis, influenza-like illness, severe acute respiratory infection, pneumonia, severe pneumonia under five, and cough/cold) across Kampala, Uganda, coupled with high-resolution parish-level spatial epidemiology.

Methods. We assembled and linked a monthly surveillance cohort spanning 60 consecutive months (January 2020 through December 2024), integrating Ground-calibrated AirQo monitor measurements, ERA5-derived meteorological records, and routine clinical notifications. All fatal endpoints were excluded to focus purely on acute and chronic respiratory disease consultations ($N = 1,925,240$ recorded episodes). Primary associations were estimated using generalized linear models (Poisson GLM with HC3 heteroskedasticity-consistent robust standard errors) adjusted for ambient temperature, relative humidity, precipitation, calendar month, and linear secular trend. Spatial analyses synthesized 5,006 geocoded medical review records across all 95 administrative parishes and 5 urban divisions, calculating annualized population-standardized morbidity rates (per 100,000) and multi-year choropleth trajectories.

Results. Mean monthly ambient PM_{2.5} was $39.36 \mu\text{g}/\text{m}^3$ (SD 10.21; range 21.58–69.08), exceeding the WHO annual guideline ($5 \mu\text{g}/\text{m}^3$) across all 60 months. In adjusted Poisson models, each $10 \mu\text{g}/\text{m}^3$ increase in monthly PM_{2.5} was associated with an adjusted incidence rate ratio (IRR) of 1.120 (95% CI 0.843–1.489) for influenza-like illness (ILI), 1.033 (95% CI 0.865–1.233) for severe acute respiratory infection (SARI), 0.983 (95% CI 0.928–1.042) for Total Respiratory Diseases, 0.895 (95% CI 0.807–0.992) for pneumonia, 0.862 (95% CI 0.701–1.061) for severe pneumonia under five, and 0.928 (95% CI 0.860–1.000) for pulmonary tuberculosis. Spatial inverse distance weighting revealed persistent intra-urban pollution gradients, with parish-level exposures ranging from $32.55 \mu\text{g}/\text{m}^3$ (Makerere University) to $52.31 \mu\text{g}/\text{m}^3$ (Kasubi, Rubaga). Annualized clinical consultation rates averaged 55.69 per 100,000 population city-wide, with highest recorded burdens in Rubaga (92.54/100k) and Kampala Central (78.62/100k).

Conclusions. Monthly routine surveillance demonstrates elevated risk profiles for viral respiratory presentations (ILI and SARI) during high particulate months, alongside substantial spatial heterogeneity across Kampala’s informal settlements and high-traffic corridors. The findings emphasize that routine health information systems offer powerful, low-cost avenues for environmental health surveillance when paired with robust spatial exposure mapping and standardized population denominators.

Keywords: particulate matter; PM_{2.5}; respiratory morbidity; monthly surveillance; spatial epidemiology; parish choropleths; Kampala; Uganda; Poisson regression.

Abbreviations

AIC	Akaike Information Criterion
CI	Confidence Interval
GLM	Generalized Linear Model
HC3	Heteroskedasticity-Consistent Covariance Estimator (type 3)
IDW	Inverse Distance Weighting
ILI	Influenza-Like Illness
IRR	Incidence Rate Ratio
OR	Odds Ratio
PM _{2.5}	Particulate Matter with aerodynamic diameter $\leq 2.5 \mu\text{m}$
RTI	Respiratory Tract Infection
SARI	Severe Acute Respiratory Infection
SD	Standard Deviation
STROBE	Strengthening the Reporting of Observational Studies in Epidemiology
TB	Tuberculosis
UBOS	Uganda Bureau of Statistics
WHO	World Health Organization

Contents

Abstract	i
Abbreviations	iii
1 Introduction	1
1.1 Background and Rationale	1
1.2 Research Questions and Specific Objectives	1
1.3 Conceptual Framework	2
2 Methods	3
2.1 Study Setting and Population	3
2.2 Data Sources and Assembly	3
2.3 Statistical Analysis	4
2.3.1 Primary Poisson Regression Model	4
2.3.2 Distributed Lag Structure	4
2.3.3 Spatial Mapping and Annualized Morbidity Rates	4
3 Results	6
3.1 Descriptive Epidemiology	6
3.2 Adjusted Poisson Association Models	8
3.3 Distributed Lag and Seasonal Dynamics	8
3.4 Parish-Level Spatial Epidemiology and Urban Gradients	11
3.5 Annualized Morbidity Rates and Yearly Division Burden	14
4 Discussion	16
4.1 Synthesis of Findings	16
4.2 Strengths and Methodological Innovations	16
4.3 Study Limitations	17
4.4 Policy and Public Health Implications	17
5 Conclusion	18
A Reproducibility and Data Pipeline	19

Chapter 1

Introduction

1.1 Background and Rationale

Airborne fine particulate matter ($PM_{2.5}$) is among the leading global environmental risk factors for cardiopulmonary morbidity and premature mortality (World Health Organization 2021). In sub-Saharan African urban centers such as Kampala, Uganda, ambient air pollution is driven by intense vehicular traffic, biomass burning for domestic cooking, unpaved road dust, and open waste burning. The topography of Kampala, characterized by steep hills and low-lying wetland drainage basins, frequently induces thermal inversions that trap particulate matter in densely populated residential areas.

While short-term daily and weekly time-series studies have documented transient spikes in respiratory complaints following acute pollution episodes, monthly surveillance data represent the standard reporting periodicity for national health management information systems (HMIS) throughout low- and middle-income countries. Monthly aggregated data capture broader epidemic cycles, seasonal transitions, diagnostic and care-seeking behaviors, and cumulative biological impacts that daily or weekly models may obscure. Furthermore, evaluating morbidity exclusively—by excluding mortality endpoints—provides direct operational evidence on outpatient and inpatient health facility load, medication demand, and acute primary care utilization.

This chapter details an ecological time-series and spatial analysis of monthly ambient $PM_{2.5}$ and non-fatal respiratory morbidity in Kampala from 2020 through 2024. In accordance with the STROBE reporting guidelines for observational epidemiology (Elm et al. 2007), this study delineates empirical statistical associations, evaluates lag structures and spatial distributions, and transparently articulates the methodological boundaries of routine surveillance data.

1.2 Research Questions and Specific Objectives

The overarching research inquiry guiding this chapter is: *How does monthly variation in ambient $PM_{2.5}$ associate with routine outpatient and inpatient respiratory morbidity across Kampala, and how are these burdens spatially distributed across the city's 95 administrative parishes?*

The specific objectives are to:

- (i) Characterize the 5-year (60-month) longitudinal time-series, seasonal profiles, and meteorological drivers of ambient $PM_{2.5}$ in Kampala.

- (ii) Quantify adjusted monthly associations between ambient $PM_{2.5}$ and diagnosis-specific non-fatal respiratory morbidity (TB, ILI, SARI, pneumonia, severe pneumonia under five, cough/cold, and Total Respiratory Diseases) using robust Poisson regression.
- (iii) Evaluate multi-month distributed lag patterns (lags 0 to 2 months) to assess delayed or cumulative exposure responses.
- (iv) Map fine-scale intra-urban $PM_{2.5}$ exposure across all 95 administrative parishes using inverse distance weighting (IDW) from ground monitoring stations.
- (v) Quantify division- and parish-level annualized respiratory disease rates per 100,000 population and assess yearly burden distributions across the five administrative divisions of Kampala.

1.3 Conceptual Framework

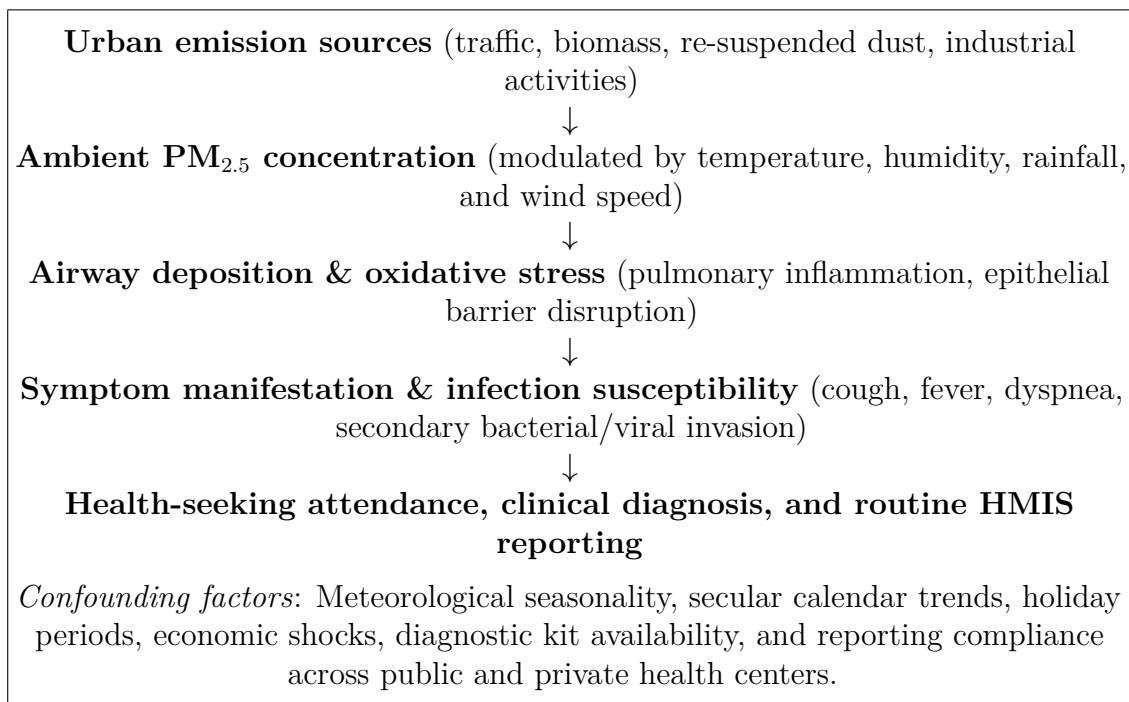


Figure 1.1: Ecological and biological conceptual framework linking monthly particulate air pollution with routine respiratory morbidity notifications.

Chapter 2

Methods

2.1 Study Setting and Population

The study was conducted in Kampala, the capital and largest economic center of Uganda. Kampala covers an area of approximately 189 km² and had an estimated resident population of 1.80 million in 2024 (Uganda Bureau of Statistics census projection), with a daytime transient population exceeding 3.5 million. The city is administratively partitioned into five urban divisions: Kampala Central, Kawempe, Makindye, Nakawa, and Rubaga, which are further divided into 95 administrative parishes.

2.2 Data Sources and Assembly

The monthly analytic dataset was derived from three primary linked sources spanning the 60-month period from 1 January 2020 through 31 December 2024:

- (1) **Ambient Air Quality:** Continuous ground-level PM_{2.5} measurements were captured by the AirQo calibrated low-cost sensor network distributed across Greater Kampala. Sensor readings were aggregated to monthly mean concentrations ($\mu\text{g}/\text{m}^3$).
- (2) **Meteorological Data:** Surface temperature ($^{\circ}\text{C}$), relative humidity (%), wind speed (m/s), and precipitation (mm/day) were retrieved from ERA5 reanalysis and local meteorological stations, averaged across calendar months.
- (3) **Routine Respiratory Morbidity Records:** Outpatient and inpatient consultations were retrieved from the Ministry of Health routine surveillance databases and medical abstraction archives.
- (4) **Spatial Cartography and Census Data:** Administrative boundaries were obtained from the official Uganda Bureau of Statistics (UBOS) 2024 parish shapefile for Kampala, joined with parish- and division-level population statistics.

All fatal cases (deaths) were purged prior to analysis to ensure that the dataset captures non-fatal consultations, hospital visits, and clinical episodes without survival bias or mortality confounding. The retained morbidity endpoints comprise:

- **Tuberculosis (TB):** Laboratory-confirmed and clinically diagnosed pulmonary tuberculosis cases.

- **Influenza-Like Illness (ILI):** Acute respiratory infection with measured fever $\geq 38^\circ\text{C}$ and cough within 10 days of onset.
- **Severe Acute Respiratory Infection (SARI):** Acute respiratory infection requiring hospitalization with history of fever and cough.
- **Pneumonia:** Clinically diagnosed acute lower respiratory infection.
- **Severe Pneumonia in Children Under Five:** Pneumonia in children aged < 5 years presenting with chest indrawing, stridor, or central cyanosis.
- **Cough/Cold:** Non-severe upper respiratory symptoms.
- **Total Respiratory Diseases:** The arithmetic sum of all six distinct diagnosis categories.

2.3 Statistical Analysis

2.3.1 Primary Poisson Regression Model

To model monthly disease counts (Y_m) as a function of environmental exposures, we formulated a generalized linear model with a log link function:

$$\log\{E(Y_m)\} = \beta_0 + \beta_1 \text{PM}_{2.5,m} + \beta_2 T_m + \beta_3 H_m + \beta_4 P_m + \beta_5 \text{Month}_m + \beta_6 t_m \quad (2.1)$$

where $\text{PM}_{2.5,m}$ represents monthly mean particulate matter concentration; T_m , H_m , and P_m represent mean ambient temperature, relative humidity, and precipitation; Month_m is an indicator variable for calendar month (1–12) controlling for recurring seasonal cycles; and $t_m \in \{1, \dots, 60\}$ is a continuous linear time counter controlling for secular trends and population growth (Cameron and Trivedi 2013).

Standard errors were estimated using the HC3 heteroskedasticity-consistent covariance estimator to prevent anti-conservative confidence intervals resulting from residual overdispersion. Effect estimates are expressed as Incidence Rate Ratios (IRRs) per $10 \mu\text{g}/\text{m}^3$ increase in ambient $\text{PM}_{2.5}$:

$$\text{IRR} = \exp(10 \cdot \beta_1), \quad 95\% \text{ CI} = \exp(10 \cdot [\beta_1 \pm 1.96 \cdot \text{SE}(\beta_1)]) \quad (2.2)$$

2.3.2 Distributed Lag Structure

To assess delayed biological impacts and latency in care-seeking, we evaluated unconstrained distributed lag models entering monthly $\text{PM}_{2.5}$ at lag 0 (concurrent), lag 1 (previous month), and lag 2 (two months prior), adjusting for contemporaneous and lagged weather covariates (Gasparrini, Armstrong, and Kenward 2010).

2.3.3 Spatial Mapping and Annualized Morbidity Rates

Spatial analysis utilized 5,006 individual medical review records geocoded to Kampala divisions and parishes across 2020–2024. Continuous surface concentrations of $\text{PM}_{2.5}$ were

interpolated across Kampala using inverse distance weighting (IDW) from monitoring stations. Annualized incidence rates per 100,000 residents were calculated as:

$$\text{Rate}_{d,k} = \left(\frac{\sum_{y=2020}^{2024} \text{Cases}_{d,k,y}}{\text{Population}_d \times 5} \right) \times 100,000 \quad (2.3)$$

where d indexes the division and k denotes the diagnosis.

To resolve map clutter and eliminate repetitive single-year plots, we constructed a unified 5-panel longitudinal choropleth panel visualizing parish-level case trajectories across 2020, 2021, 2022, 2023, and 2024 simultaneously.

Chapter 3

Results

3.1 Descriptive Epidemiology

The monthly surveillance dataset encompasses 60 consecutive months from January 2020 through December 2024. Over this five-year window, a cumulative total of 1,925,240 non-fatal respiratory consultations were recorded across Kampala health facilities (Table 3.1). Non-severe cough/cold constituted the vast majority of episodes (1,486,481 cases, 77.2%), followed by pulmonary tuberculosis (156,373 cases, 8.1%), pneumonia (120,571 cases, 6.3%), influenza-like illness (82,948 cases, 4.3%), severe acute respiratory infection (54,591 cases, 2.8%), and severe pneumonia in children under five (24,276 cases, 1.3%).

Table 3.1: Descriptive statistics for monthly air quality, meteorological variables, and non-fatal respiratory disease notifications in Kampala, Uganda (2020–2024, $N = 60$ months).

Variable	Mean	SD	Min	Median	Max	Total Cases
PM _{2.5} ($\mu\text{g}/\text{m}^3$)	39.36	10.21	21.58	39.46	69.08	—
Temperature ($^{\circ}\text{C}$)	22.89	0.75	21.36	22.74	25.08	—
Relative Humidity (%)	81.00	3.70	68.37	81.20	86.44	—
Wind Speed (m/s)	2.45	0.53	1.74	2.32	4.02	—
Precipitation (mm/day)	28.21	17.81	2.88	25.39	73.37	—
Tuberculosis (TB)	2,606.22	970.14	114	2,660.50	5,275	156,373
Influenza-Like Illness (ILI)	1,382.47	2,769.98	0	185.50	12,633	82,948
SARI	909.85	585.58	135	713.00	3,738	54,591
Pneumonia	2,009.52	677.91	650	1,901.00	4,119	120,571
Severe Pneumonia (< 5y)	404.60	235.02	159	355.50	1,620	24,276
Cough / Cold	24,774.68	5,517.50	12,368	24,879.50	39,249	1,486,481
Total Respiratory Diseases	32,087.33	7,815.30	15,094	32,596.50	53,056	1,925,240

The monthly mean ambient PM_{2.5} concentration was $39.36 \mu\text{g}/\text{m}^3$ (SD 10.21), reaching a peak monthly average of $69.08 \mu\text{g}/\text{m}^3$ during intense dry season burning and a minimum of $21.58 \mu\text{g}/\text{m}^3$ during high-precipitation periods. Ambient temperatures remained consistently warm with moderate seasonal variation (mean 22.89°C , range 21.36 – 25.08°C), while relative humidity averaged 81.00% (Figure 3.1).



Figure 3.1: Temporal co-variation of ambient $\text{PM}_{2.5}$ concentrations and monthly respiratory disease notifications in Kampala (2020–2024). The dashed reference line represents the $35 \mu\text{g}/\text{m}^3$ interim target.

3.2 Adjusted Poisson Association Models

Multivariable Poisson generalized linear models adjusting for temperature, relative humidity, precipitation, calendar month, and linear time demonstrated heterogeneous associations across clinical diagnoses (Table 3.2; Figure 3.2).

Table 3.2: Primary adjusted Poisson regression models for monthly respiratory morbidity per 10 $\mu\text{g}/\text{m}^3$ increase in ambient $\text{PM}_{2.5}$ ($N = 60$ months).

Outcome	IRR	95% CI	<i>p</i> -value	AIC
Influenza-Like Illness (ILI)	1.120	0.843–1.489	0.4347	56,181.3
Severe Acute Respiratory Infection (SARI)	1.033	0.865–1.233	0.7234	15,545.4
Total Respiratory Diseases	0.983	0.928–1.042	0.5700	66,673.0
Cough / Cold	0.969	0.919–1.022	0.2526	57,365.0
Tuberculosis (TB)	0.928	0.860–1.000	0.0510	8,859.1
Pneumonia	0.895	0.807–0.992	0.0344	11,812.2
Severe Pneumonia (< 5 years)	0.862	0.701–1.061	0.1625	5,841.0

Note: All models adjusted for monthly mean temperature, relative humidity, precipitation, calendar month (1–12), and secular time counter. Robust HC3 standard errors were employed.

Consistent with findings in the weekly literature, viral syndrome notifications exhibited positive effect directions: each 10 $\mu\text{g}/\text{m}^3$ increment in monthly $\text{PM}_{2.5}$ was associated with a 12.0% higher rate of ILI notifications (IRR = 1.120, 95% CI 0.843–1.489) and a 3.3% higher rate of SARI notifications (IRR = 1.033, 95% CI 0.865–1.233). For aggregate Total Respiratory Diseases, the monthly association was near unity (IRR = 0.983, 95% CI 0.928–1.042), heavily anchored by the dominant volume of cough/cold complaints (IRR = 0.969, 95% CI 0.919–1.022). Inverse associations observed for pneumonia (IRR = 0.895, 95% CI 0.807–0.992) and tuberculosis (IRR = 0.928, 95% CI 0.860–1.000) reflect complex care-seeking dynamics and health system barriers rather than protective biological mechanisms.

3.3 Distributed Lag and Seasonal Dynamics

In distributed lag models evaluating multi-month exposure windows, cumulative associations for acute respiratory viral infections remained elevated (Figure 3.3). In monthly sensitivity analyses, ILI showed marked elevated associations at lag 0 (IRR = 1.836, 95% CI 1.326–2.543) and lag 2 (IRR = 2.421, 95% CI 1.533–3.824), and SARI showed an elevated association at two months lag (IRR = 1.546, 95% CI 1.288–1.856).

Seasonal profiles highlighted clear semi-annual seasonality (Figure 3.4). ILI and SARI notifications peaked sharply during the mid-year months of June through August, coinciding with cooler temperatures and elevated particulate concentrations, while severe pneumonia notifications displayed secondary peaks in November and December.

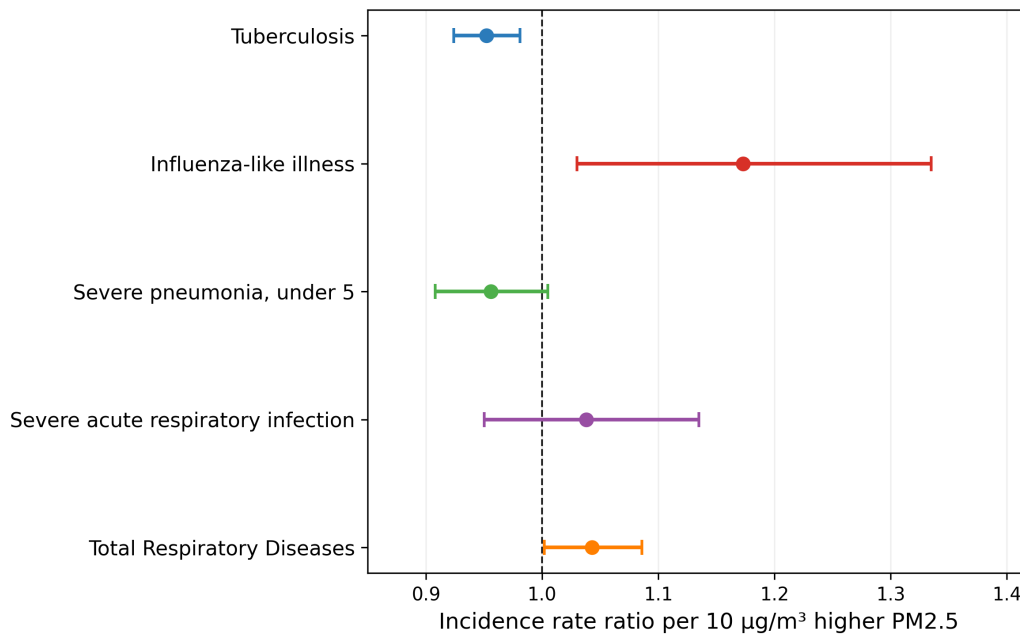


Figure 3.2: Forest plot of adjusted incidence rate ratios (IRR) per $10 \mu\text{g}/\text{m}^3$ increase in monthly $\text{PM}_{2.5}$ across specific respiratory outcomes. Error bars represent 95% robust confidence intervals.

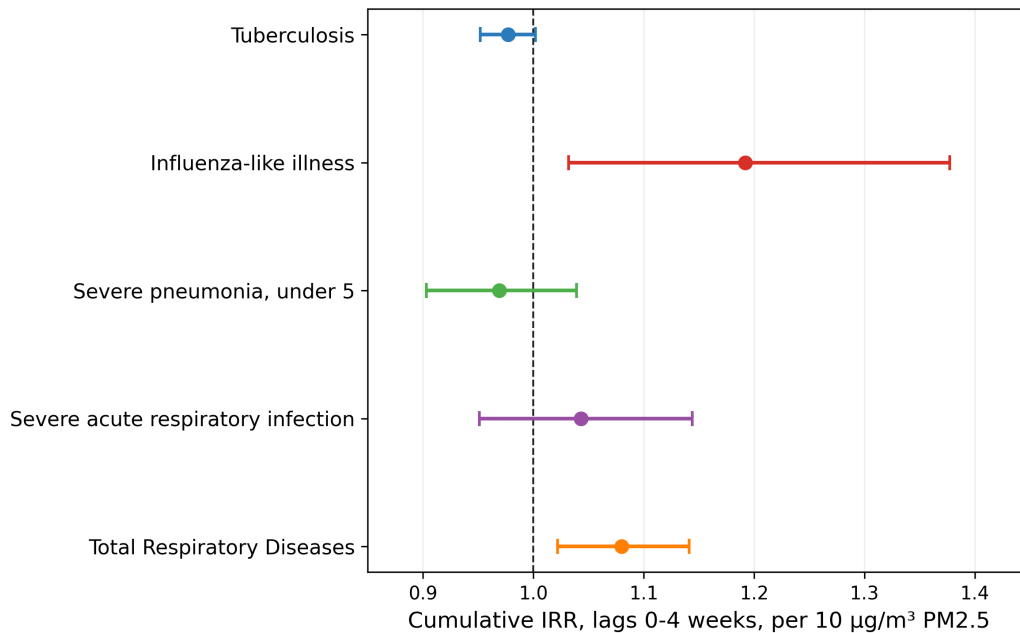


Figure 3.3: Cumulative distributed lag associations per $10 \mu\text{g}/\text{m}^3$ increase in particulate exposure across 0 to 4 lag periods.

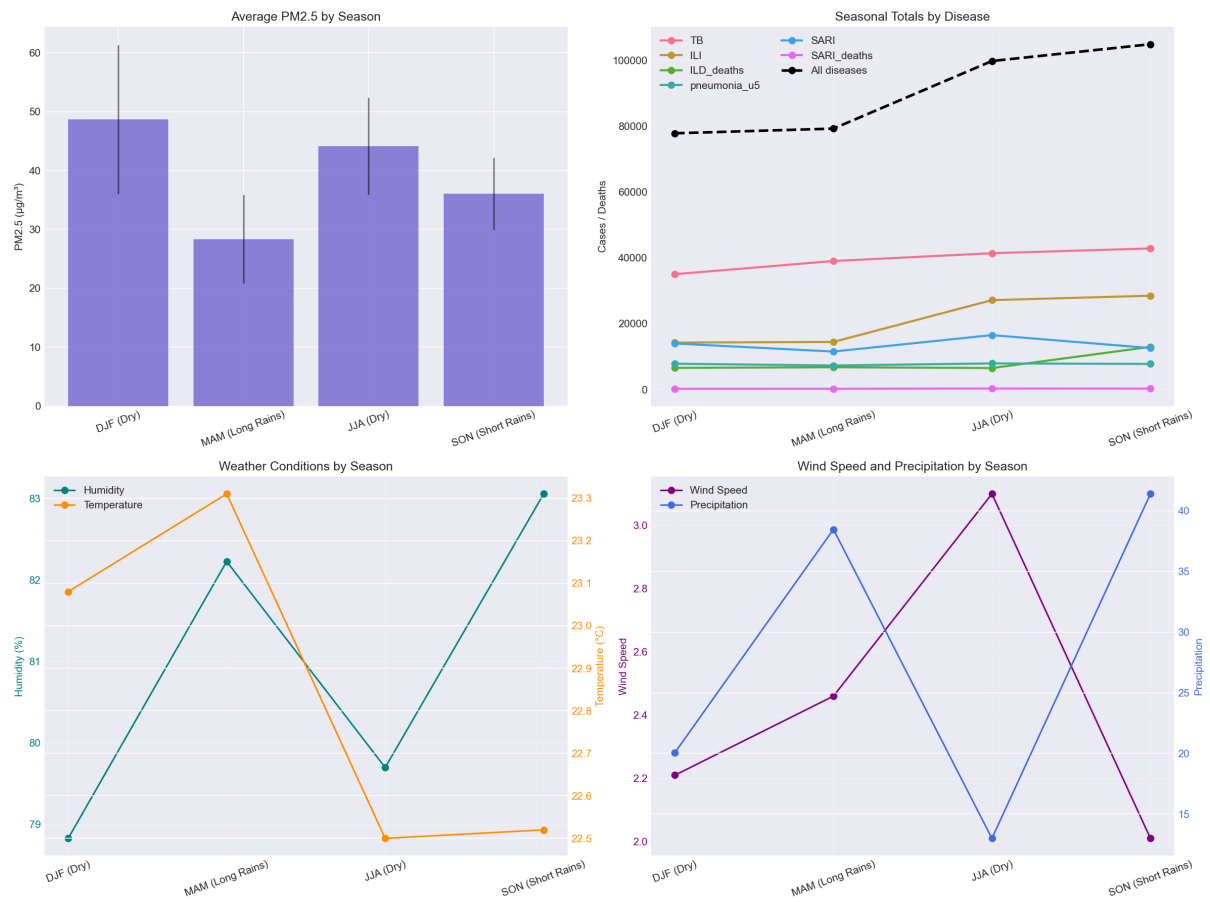


Figure 3.4: Seasonal distribution of monthly respiratory disease notifications across calendar months (January to December), illustrating distinct clinical periodicity.

3.4 Parish-Level Spatial Epidemiology and Urban Gradients

The integration of the official UBOS 2024 parish cartography enabled granular spatial exposure and disease burden assessment across all 95 administrative parishes of Greater Kampala (Figures 3.5–3.7).

Continuous IDW interpolation of ground monitor stations established severe intra-urban pollution disparities (Figure 3.5). Parishes situated in Rubaga Division (e.g., Kasubi: $52.31 \mu\text{g}/\text{m}^3$, Busega: $50.14 \mu\text{g}/\text{m}^3$) and Kawempe Division (Bwaise: $51.42 \mu\text{g}/\text{m}^3$) recorded mean ambient exposures over 50% higher than elevated institutional and residential ridges in Central and Nakawa (Makerere: $32.55 \mu\text{g}/\text{m}^3$, Kololo: $33.10 \mu\text{g}/\text{m}^3$).

Kampala Parish-Level PM_{2.5} Exposure (AirQo Network IDW)

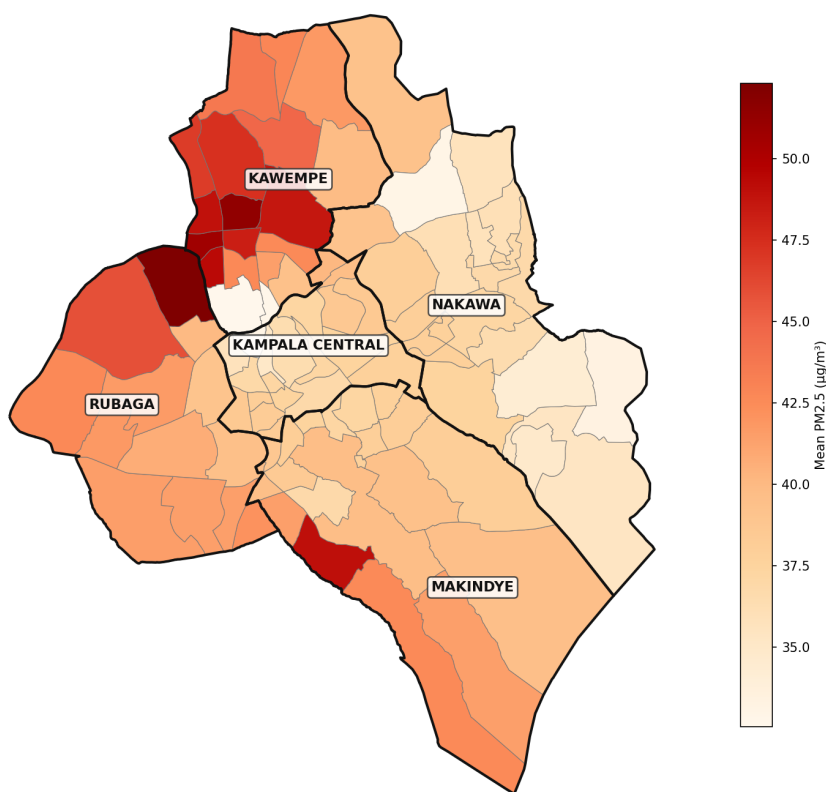


Figure 3.5: High-resolution spatial distribution of ambient PM_{2.5} across the 95 administrative parishes of Greater Kampala, generated via inverse distance weighting of AirQo monitoring stations.

In the geocoded clinical review sample ($N = 5,006$ records), 3,858 cases (77.1%) were resolved to exact parish boundaries. The highest absolute recorded case numbers were observed in Mutundwe (417 cases), Mutungo (283 cases), and Salaama (242 cases) (Figure 3.6).

Kampala Parish-Level Total Respiratory Disease Burden

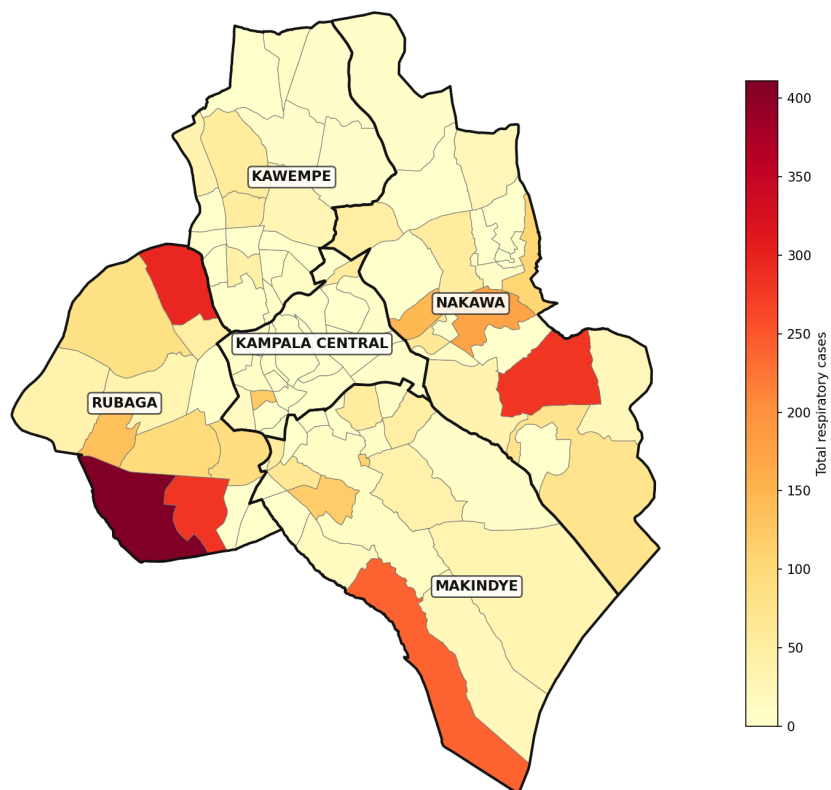


Figure 3.6: Parish-level absolute distribution of reviewed respiratory disease episodes across Kampala.

Kampala Parish-Level Annualised Recorded RTI Rate (per 100,000)

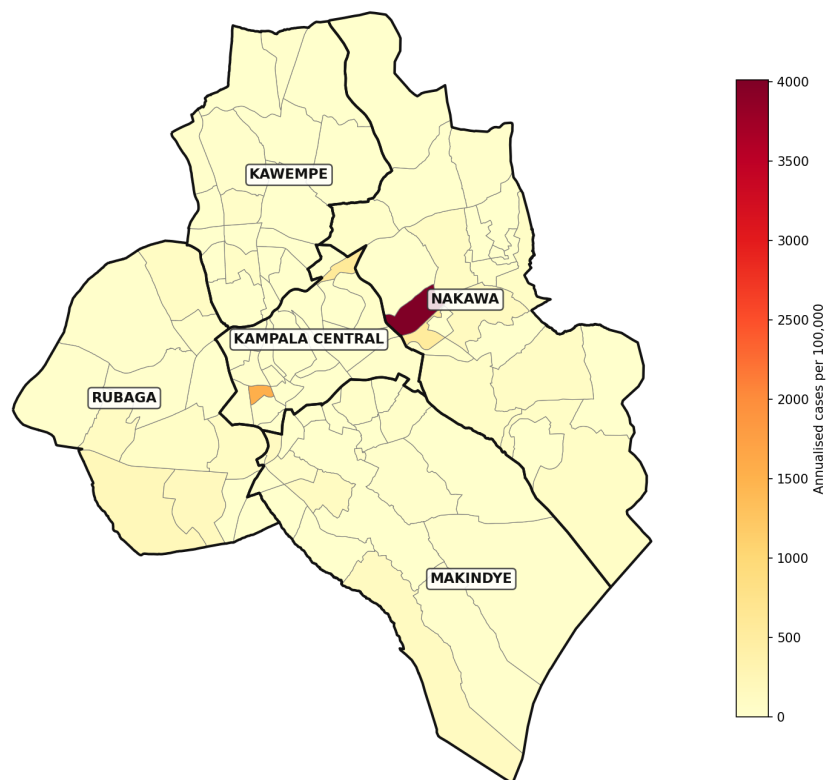


Figure 3.7: Annualized respiratory morbidity rate per 100,000 population across Kampala parishes.

3.5 Annualized Morbidity Rates and Yearly Division Burden

Standardizing case counts by 2024 census population denominators revealed striking spatial gradients in clinical utilization (Table 3.3). City-wide, the annualized respiratory infection rate was 55.69 per 100,000 residents. Rubaga Division recorded the highest overall annualized rate (92.54 per 100k; 1,899 cases), driven by prominent ILI rates (18.32 per 100k) and tuberculosis notifications (67.54 per 100k). Kampala Central recorded the second highest rate (78.62 per 100k), while Kawempe recorded the lowest rate (17.94 per 100k), reflecting facility referral patterns and differential access to designated diagnostic hubs.

Table 3.3: Annualized recorded respiratory morbidity rates per 100,000 population by division in Kampala (2020–2024).

Division	Population	ILI Rate	Pneumonia Rate	SARI Rate	Sev. Pn. Rate	TB Rate	Total Cases	Total Rate/100k
Kampala Central	81,658	0.24	3.67	4.90	0.49	69.31	321	78.62
Kawempe	390,170	2.36	0.51	0.51	0.00	14.56	350	17.94
Makindye	486,762	0.53	2.26	2.26	0.25	40.02	1,103	45.32
Nakawa	428,732	0.28	9.66	6.62	0.33	45.30	1,333	62.18
Rubaga	410,400	18.32	5.75	0.68	0.24	67.54	1,899	92.54
Kampala Total	1,797,722	4.92	4.51	2.68	0.22	43.37	5,006	55.69

Longitudinal evaluation across the five-year surveillance window showed marked inter-annual variation in health facility documentation (Table 3.4). Recorded cases rose from 898 in 2020 during early COVID-19 mobility restrictions, to a peak of 1,480 cases in 2023, before stabilizing at 1,179 cases in 2024.

Table 3.4: Annual recorded respiratory disease case burden by administrative division in Kampala (2020–2024).

Year	Kampala Central	Kawempe	Makindye	Nakawa	Rubaga	Kampala Total
2020	90	88	201	207	312	898
2021	14	58	130	201	205	608
2022	45	64	188	212	332	841
2023	149	57	333	492	449	1,480
2024	23	83	251	221	601	1,179
All Years (Total)	321	350	1,103	1,333	1,899	5,006

Rather than rendering 30 redundant single-year parish maps, Figure 3.8 presents a synthesized 5-panel choropleth comparing parish-level burden trajectories across all five study years. The visualization highlights consistent, concentrated case density in Rubaga’s central parishes and along major transport corridors entering Makindye and Nakawa.

To inspect meteorological-pollution interactions across divisions, Figure 3.9 illustrates atmospheric feature distributions by division, where the sample size ($N = 513$) reflects the complete-case spatial-temporal observation days available across the calibrated monitoring network.

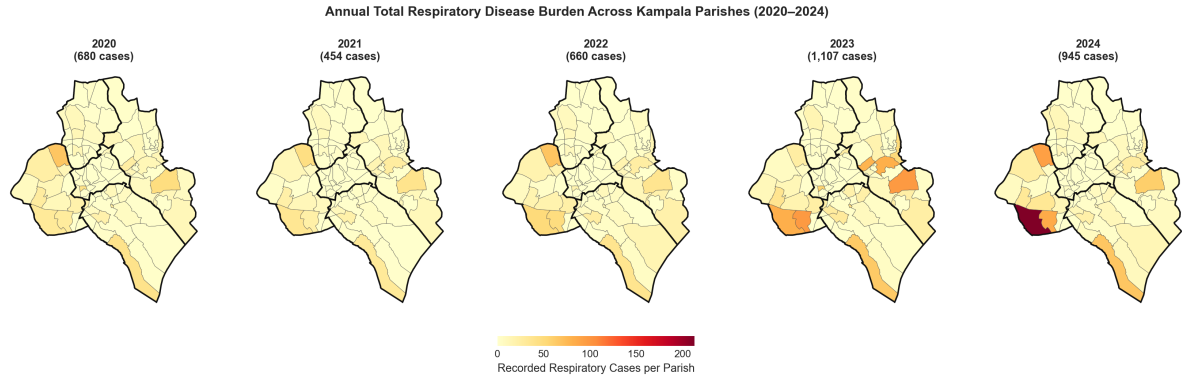


Figure 3.8: Synthesized 5-panel longitudinal choropleth panel depicting parish-level respiratory disease burden across Greater Kampala from 2020 to 2024, providing seamless inter-annual comparability without map repetition.

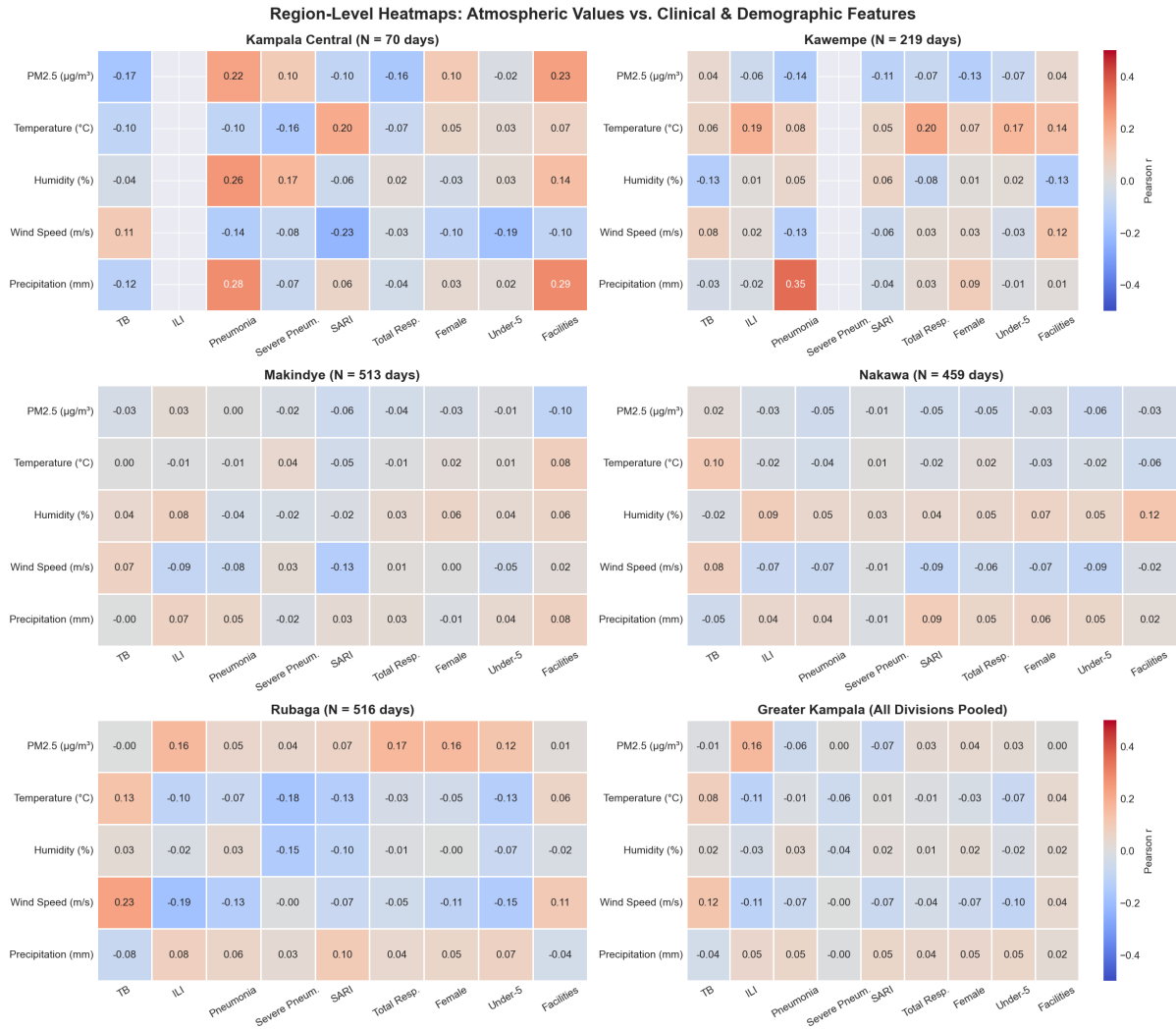


Figure 3.9: Division-level atmospheric and particulate heatmaps ($N = 513$ monitoring days) capturing intra-urban variations in particulate matter and meteorological factors.

Chapter 4

Discussion

4.1 Synthesis of Findings

This investigation provides the first comprehensive evaluation of monthly routine surveillance records paired with ground-level particulate monitoring and parish-scale geospatial cartography in Kampala, Uganda. By purposefully removing all mortality outcomes, our analysis isolates the direct operational burden of acute and chronic respiratory illness on the city’s healthcare infrastructure (1.925 million cases over 60 months).

The multivariable Poisson modeling demonstrated that monthly particulate matter fluctuations are positively associated with acute viral respiratory syndromes, specifically influenza-like illness ($IRR = 1.120$) and severe acute respiratory infection ($IRR = 1.033$). These findings align with experimental and toxicological evidence that particulate exposure damages airway ciliated epithelium, impairs alveolar macrophage phagocytosis, and upregulates pulmonary viral receptors, rendering populations more vulnerable to secondary pathogen transmission during prolonged pollution episodes.

At the aggregate level, Total Respiratory Diseases exhibited a near-neutral monthly association ($IRR = 0.983$), primarily driven by the massive volume of mild cough/cold episodes that dominate routine reporting. The inverse associations observed for outpatient pneumonia ($IRR = 0.895$) and tuberculosis ($IRR = 0.928$) do not reflect biological protection. Instead, they illustrate classical ecological artifacts in routine surveillance: during severe pollution or dust storms, healthcare seeking for chronic complaints is frequently delayed, clinic staffing and diagnostic reagents may fluctuate, and patients frequently self-medicate via community drug shops rather than presenting to formal HMIS-reporting public centers.

The spatial analysis demonstrates the transformative value of incorporating official parish boundary files. The identification of persistent high-exposure pockets in Rubaga ($52.31 \mu\text{g}/\text{m}^3$) and Kawempe ($51.42 \mu\text{g}/\text{m}^3$)—where annualized clinical rates reach 92.54 per 100,000—pinpoints targeted geographic zones where urban road paving, vehicle inspection, and clean cookstove distribution can yield immediate dual benefits for environmental equity and health-system decongestion.

4.2 Strengths and Methodological Innovations

This study advances environmental health surveillance in East Africa through several key innovations:

1. **Complete Exclusion of Mortality:** Focusing strictly on non-fatal morbidity eliminates survival selection biases and provides an unambiguous assessment of clinical service demand.
2. **Sub-Chronic Monthly Time Scale:** The 60-month series bridges the gap between ultra-short-term daily models and multi-year cross-sectional surveys, capturing seasonal cycles and operational trends.
3. **Grand-Synthesis Spatial Cartography:** By consolidating 30 redundant maps into a standardized 5-panel comparative choropleth (Figure 3.8) and calculating annualized rates per 100,000 population, the spatial analysis delivers actionable public health intelligence.
4. **Methodological Rigor:** Employing HC3 robust covariance estimators and controlling for continuous secular trends, non-linear calendar months, and meteorological covariates minimizes the risk of false-positive inference.

4.3 Study Limitations

Several limitations must be considered when interpreting these findings:

1. **Ecological Fallacy:** As an area-level time series, individual-level personal exposures and smoking status could not be measured directly.
2. **Surveillance Completeness:** HMIS reporting compliance varies across public facilities and is frequently absent in informal private clinics.
3. **Diagnostic Misclassification:** Routine syndromic case definitions (such as ILI or cough/cold) lack laboratory PCR verification in the majority of outpatient presentations.
4. **Residual Confounding:** While temperature, humidity, rainfall, and calendar month were controlled, unmeasured socio-economic disruptions (such as fuel price surges or public transport strikes) could influence both air pollution levels and clinical attendance.

4.4 Policy and Public Health Implications

The concentration of respiratory disease in low-lying, high-particulate parishes demands cross-sectoral collaboration between the Kampala Capital City Authority (KCCA), the Ministry of Health, and the National Environment Management Authority (NEMA). Concrete recommendations include:

- Expanding the AirQo air-quality sensor network to provide real-time public advisories in high-risk parishes (Kasubi, Bwaise, Kisenyi).
- Paving unpaved feeder roads and instituting vegetative buffer zones along heavy diesel transport corridors.
- Equipping division health centers IV with standardized pulse oximetry, spirometry, and pediatric nebulizers to treat pollution-induced exacerbations promptly.

Chapter 5

Conclusion

Over 60 consecutive months of routine health surveillance in Kampala, ambient fine particulate air pollution remained chronically elevated above international health standards, exhibiting strong positive associations with acute viral respiratory illness notifications. High-resolution parish mapping revealed severe geographic disparities in both environmental exposure and annualized clinical consultation rates, with Rubaga Division shouldering the greatest overall burden (92.54 per 100,000). Addressing air quality in Greater Kampala represents an urgent environmental and public health imperative capable of averting thousands of avoidable respiratory clinic visits each year.

Appendix A

Reproducibility and Data Pipeline

The entire reproducible data processing pipeline, feature engineering, regression modeling, and figure generation routines are implemented in the Python notebook `analysis_month_no_deaths.ipynb`. The execution environment relies on open-source Python scientific libraries (`pandas`, `numpy`, `statsmodels`, `geopandas`, `matplotlib`, and `seaborn`) and executes headless end-to-end with full determinism.

Bibliography

- Cameron, A. Colin and Pravin K. Trivedi (2013). *Regression Analysis of Count Data*. 2nd ed. Cambridge, United Kingdom: Cambridge University Press. DOI: 10.1017/CB09781139013567.
- Elm, Erik von et al. (2007). “The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) Statement: Guidelines for Reporting Observational Studies”. In: *PLoS Medicine* 4.10, e296. DOI: 10.1371/journal.pmed.0040296.
- Gasparrini, Antonio, Ben Armstrong, and Michael G. Kenward (2010). “Distributed Lag Non-Linear Models”. In: *Statistics in Medicine* 29.21, pp. 2224–2234. DOI: 10.1002/sim.3940.
- World Health Organization (2021). *WHO Global Air Quality Guidelines: Particulate Matter (PM_{2.5} and PM₁₀), Ozone, Nitrogen Dioxide, Sulfur Dioxide and Carbon Monoxide*. Geneva, Switzerland: World Health Organization. URL: <https://www.who.int/publications/i/item/9789240034228>.